

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS (5 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 2 – Application Based Questions
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO1 Statement:	<i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data (BL3).</i>		
Student Name:	RENEE VINCENT ASR	Reg. No.:	192424161
Section / Batch:	BTECH AI&DS	Date:	

Code of Conduct

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Amounts & Proportions	Category-wise comparison and proportion analysis using bar charts	1	5
Distributions	Analyzing data spread and patterns using histograms, density plots, boxplots	2	5
Temporal & x–y Relationships	Studying trends over time and relationships between variables	3	5
Geospatial & Temporal Data	Mapping locations and analyzing delivery patterns over time	4	5
Multivariate Data Visualization	Exploring multiple variables and comparing distributions	5	5
GRAND TOTAL:			25 Marks

Annexure A – Application Based Questions

1.A university wants to analyze student enrollments across various departments such as Computer Science, Mechanical Engineering, Civil Engineering, and Electronics. The administration aims to identify departments with the highest enrollments, compare enrollment trends over multiple semesters, and understand the proportion of students in each department. How would you categorize the data and choose appropriate visualizations to represent enrollment amounts, proportions, and temporal trends across departments?

Aim

To analyze student enrollments across different departments using suitable visualization techniques in R.

Bar Plot

- A bar plot compares student enrollment across different departments.
- Each bar represents one department.
- It helps identify departments with the highest and lowest enrollments.
- Best suited for comparing categorical data.

Pie Chart

- A pie chart shows the proportion of students enrolled in each department.
- Each slice represents a department's share of the total enrollment.
- Helps understand the percentage contribution of each department.
- Useful for visualizing part-to-whole relationships.

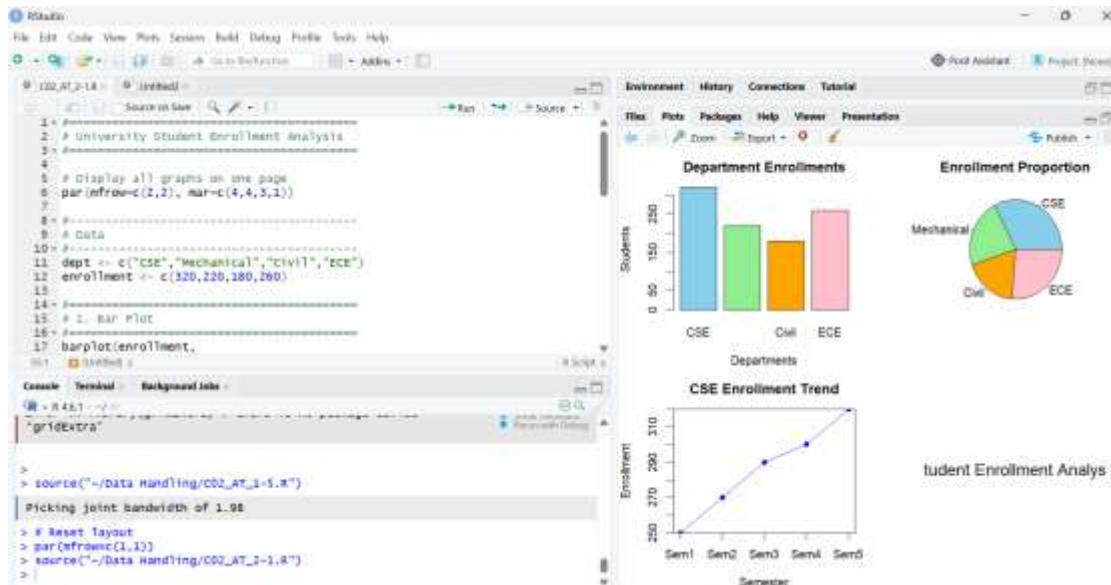
Line Chart

- A line chart shows enrollment trends across multiple semesters.
- It connects enrollment values over time.
- Helps identify increasing or decreasing enrollment patterns.
- Useful for analyzing temporal changes in student admissions.

Justification of Visualizations

Visualization	Purpose
Bar Plot	Compares enrollment across departments.
Pie Chart	Shows the proportion of students in each department.
Line Chart	Displays enrollment trends over multiple semesters.

Output



2.A manufacturing company records the number of defects found in products produced by different production lines. The quality assurance team wants to examine the distribution of defects, identify variability among production units, and detect unusual patterns. How would you use histograms, density plots, violin plots, and boxplots to analyze defect distributions and compare quality performance across production lines?

Aim

To analyze defect distributions across different production lines using suitable visualization techniques in R.

Histogram

- A histogram displays the frequency distribution of product defects.
- It groups defect counts into intervals (bins).
- Helps identify common defect ranges and overall distribution.
- Useful for understanding defect occurrence patterns.

Density Plot

- A density plot provides a smooth representation of defect distribution.
- It helps identify peaks, spread, and skewness in defect data.
- Useful for comparing overall quality performance.
- Helps detect concentration of defects.

Violin Plot

- A violin plot combines a box plot with a density plot.
- It displays the distribution and variability of defects across production lines.
- Helps compare defect patterns among production units.
- Useful for identifying production lines with inconsistent quality.

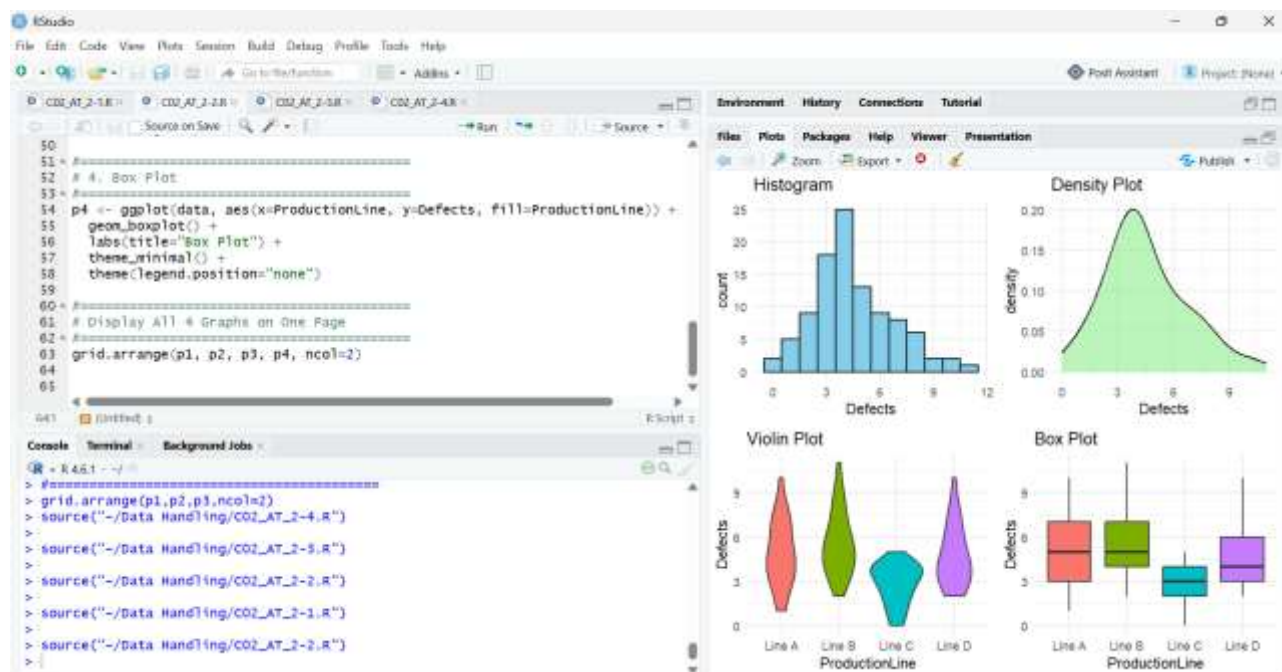
Box Plot

- A box plot summarizes defect data using the median, quartiles, and outliers.
- It helps compare variability among production lines.
- Detects unusual defect counts and quality issues.
- Useful for identifying production lines requiring improvement.

Justification of Visualizations

Visualization	Purpose
Histogram	Displays the frequency distribution of product defects.
Density Plot	Shows the overall distribution of defect counts.
Violin Plot	Compares defect distributions across production lines.
Box Plot	Identifies variability and outliers in defect data.

Output



3. An online learning platform tracks student activity including time spent on courses, number of videos watched, quiz scores, and course completion rates. The platform seeks to understand relationships among these variables and identify factors influencing successful course completion. How would you use scatter plots, heatmaps, and multivariate visualizations to analyze relationships among learner engagement metrics and academic outcomes?

Aim

To analyze learner engagement metrics and their relationship with course completion using suitable visualization techniques in R.

Scatter Plot

- A scatter plot shows the relationship between two learning variables.
- Each point represents one student.

- Helps identify positive or negative correlations between engagement metrics.
- Useful for analyzing factors affecting course completion.

Heatmap

- A heatmap displays engagement values using different color intensities.
- Darker colors indicate higher values, while lighter colors indicate lower values.
- Helps compare learner engagement across multiple metrics.
- Useful for identifying patterns in student performance.

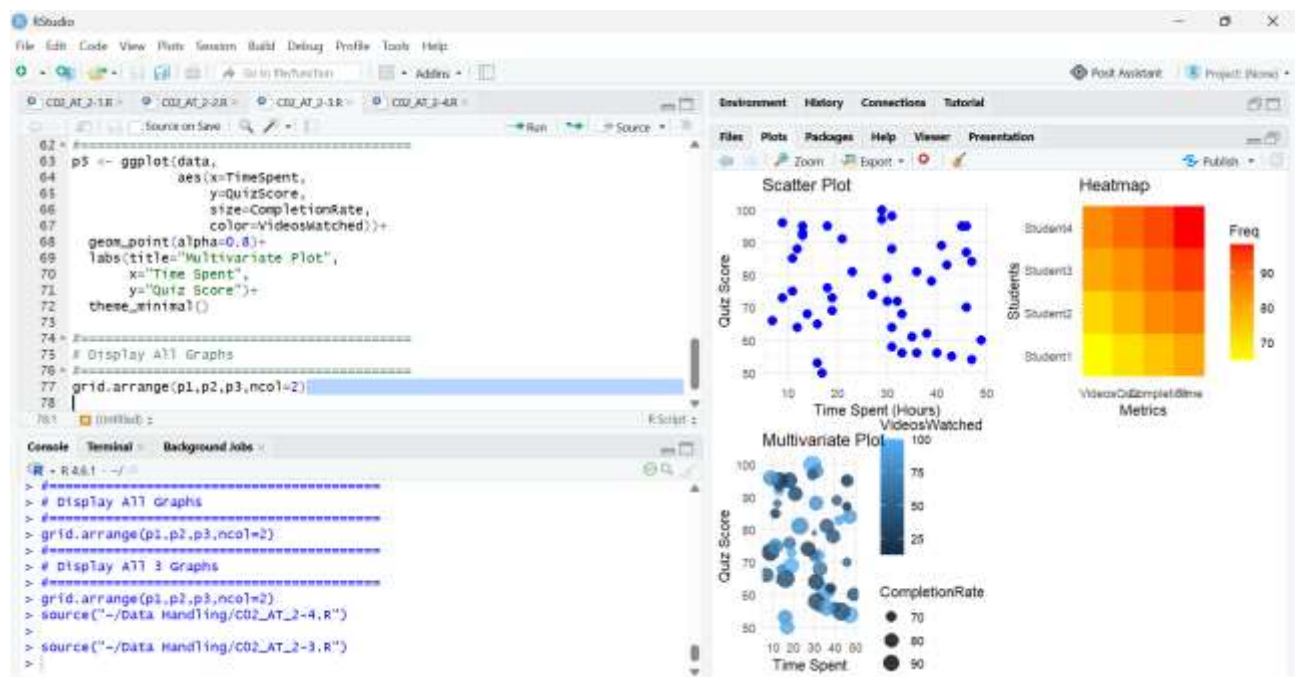
Multivariate Plot

- A multivariate plot displays multiple variables in a single graph.
- It compares time spent, quiz scores, videos watched, and course completion simultaneously.
- Helps understand relationships among several learning metrics.
- Useful for identifying key factors influencing academic success.

Justification of Visualizations

Visualization	Purpose
Scatter Plot	Shows relationships between learner engagement variables.
Heatmap	Highlights engagement intensity across different metrics.
Multivariate Plot	Compares multiple learning variables simultaneously.

Output



4.A city administration collects traffic data including vehicle counts, congestion levels, accident locations, and travel times across different regions and time periods. The objective is to identify traffic hotspots and improve transportation planning. How would

you represent geospatial and temporal data using maps, heatmaps, and time-series visualizations to identify traffic patterns and optimize road management?

Aim

To analyze traffic patterns and identify congestion hotspots using suitable visualization techniques in R.

Geospatial Map

- A geospatial map displays traffic information based on geographical locations.
- It helps visualize vehicle counts, accident locations, and congestion across different regions.
- Different colors or symbols indicate varying traffic conditions.
- Useful for identifying traffic hotspots and planning road improvements.

Heatmap

- A heatmap represents traffic congestion using different color intensities.
- Darker colors indicate higher congestion, while lighter colors indicate lower congestion.
- Helps quickly identify busy roads and peak traffic regions.
- Useful for monitoring traffic density across locations and time.

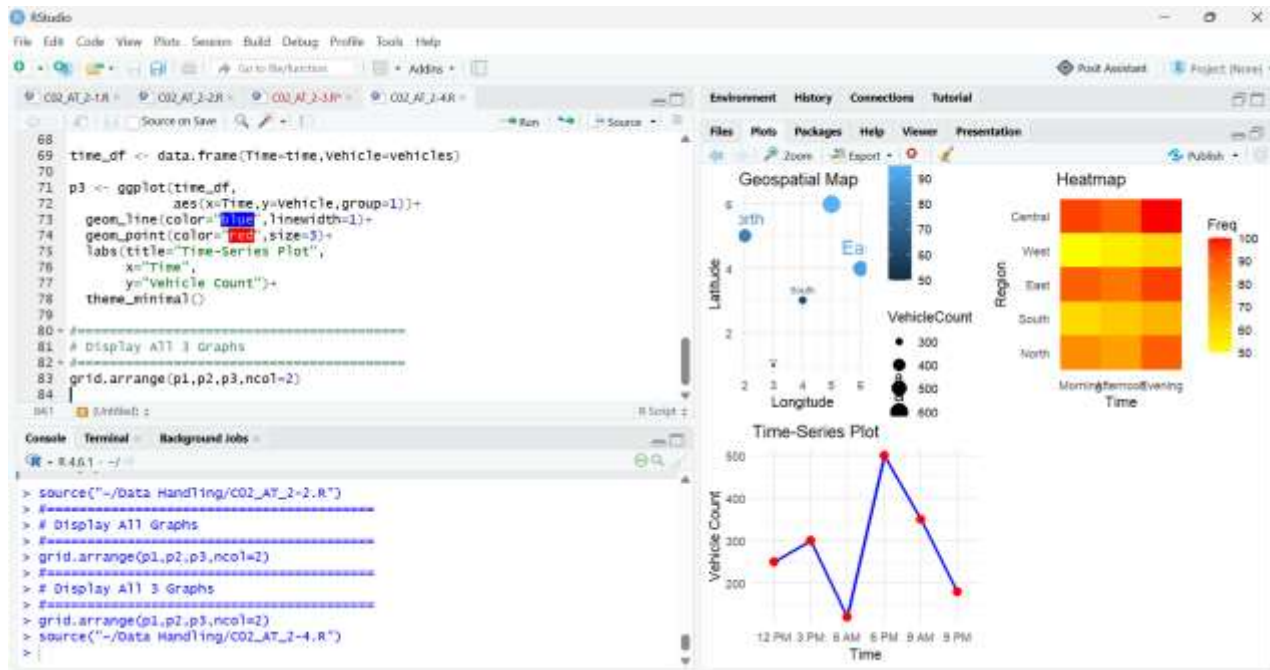
Time-Series Visualization

- A time-series plot shows changes in traffic conditions over time.
- It represents trends in vehicle counts, travel time, or congestion levels.
- Helps identify peak traffic hours and seasonal traffic patterns.
- Useful for analyzing transportation efficiency and traffic flow.

Justification of Visualizations

Visualization	Purpose
Geospatial Map	Displays traffic conditions across different regions.
Heatmap	Highlights congestion intensity across locations and time.
Time-Series Plot	Shows traffic trends and travel patterns over time.

Output



5.A large organization records communication data among employees through emails, meetings, and project collaborations. Management wants to identify key employees, collaboration groups, and communication bottlenecks within the organization. How would you use network visualizations and multivariate analysis techniques to represent communication patterns, identify influential individuals, and improve organizational collaboration?

Aim

To analyze organizational communication patterns, identify key influencers, and discover collaboration bottlenecks using network visualizations and multivariate analysis techniques in R.

Social Network Diagram

- A network graph displays employees as nodes and communication channels (emails, meetings, collaborations) as links.
- It helps visualize interactions, structural hierarchies, and isolated team members across the organization.
- Node sizes and edge weights indicate communication frequency and interaction strength.
- Useful for identifying central hubs, key information brokers, and communication bottlenecks.

Clustering / Community Detection Visualization

- A grouped network plot uses community detection algorithms to group closely interacting employees together.
- Different color clusters represent distinct cross-functional teams and collaboration groups.
- Helps reveal informal work networks versus official department boundaries.
- Useful for evaluating team cohesion and identifying cross-departmental silos.

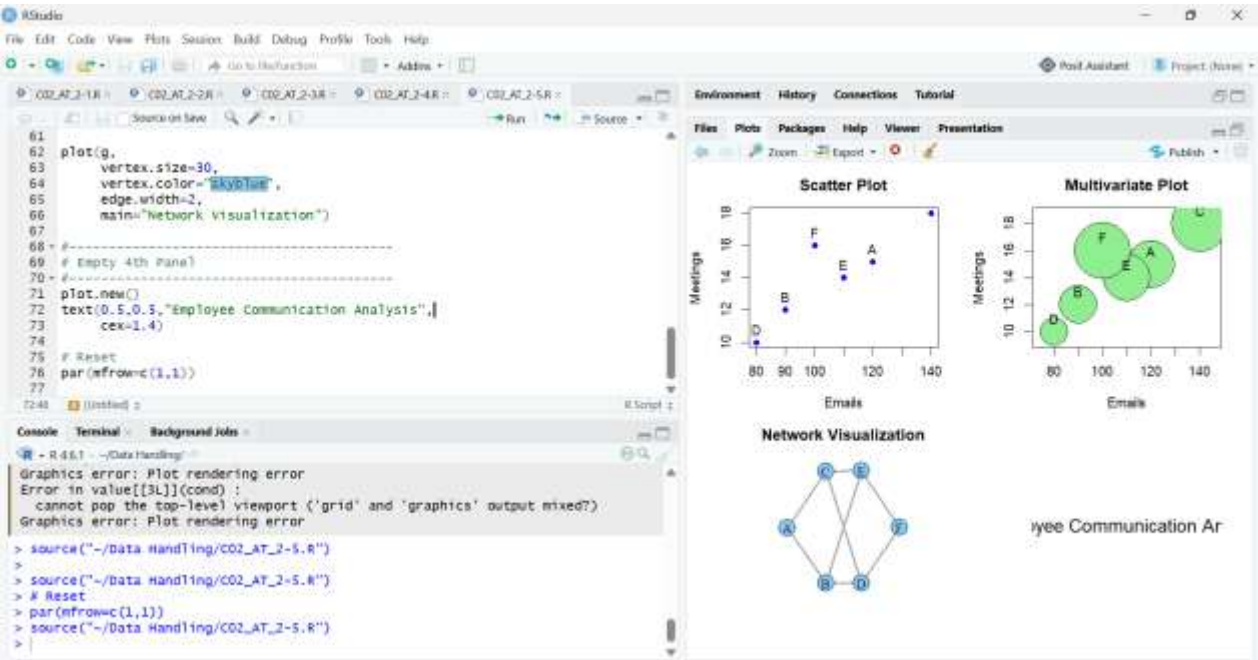
Multivariate Biplot (PCA / MCA Visualization)

- A biplot overlays employee attributes (e.g., project hours, email volume, meeting counts) alongside communication metrics.
- It projects multidimensional collaboration data onto a simple 2D or 3D coordinate space.
- Vector directions and lengths highlight key factors driving organizational communication.
- Useful for comparing employee performance profiles and understanding overall workload distribution.

Justification of Visualizations

Visualization	Purpose
Social Network Diagram	Maps employee relationships, highlighting key hubs and communication bottlenecks.
Clustering Plot	Group employees into distinct collaboration clusters to identify informal teams and organizational silos.
Multivariate Biplot	Analyzes multiple communication metrics simultaneously to evaluate overall employee influence and engagement patterns.

Output



Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem		
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts		
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach		
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution		
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____